Royal Devon University Healthcare NHS Foundation Trust

*For **clinical guidelines**, see the separate clinical guideline template*

Standard Operating Procedure for managing hypercapnic respiratory failure with Non-Invasive Ventilation (Northern Services)

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Division/ Department responsible for Procedural Document	Medical Division
Contact details	
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Monitoring Information		**Strategic Directions – Key Milestones**	
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Assurance Framework		Integrated Community Pathways	
Monitor/Finance/Performance		Develop Acute Services	
CQC Fundamental Standards Regulations No:		Delivery of Care Closer to Home	
		Infection Control	
Other *(please specify)*:			
Note: This document has been assessed for any equality, diversity or human rights implications			

Controlled document
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Full History		**Status: Draft** or **Final**	
Version	Date	Author (Title not name)	Reason
1	July 2020		*New document*
1.1	July 2023		*Updates added around EPIC, TEP considerations, patients on long term NIV and safe staffing*

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Contact for Review:	ACP Respiratory Medicine

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KEY POINTS OF THIS PROCEDURAL DOCUMENT:

1. INTRODUCTION

This Standard Operating procedure (SOP) supports the clinical management and safe staffing requirements when caring for patients requiring non-invasive ventilation across North Devon District Hospital outside of Intensive care.

2. PURPOSE

2.1. The purpose of this document is to provide clinical guidance to assist in the management of patients with hypercapnic respiratory failure who require ward based non-invasive ventilation (NIV)

2.2. The policy applies to in-patients with hypercapnic respiratory failure

2.3. Implementation of this policy will ensure that:

- Patients are safely managed and receive adequate treatment for their hypercapnic respiratory failure
- It is recommended that patients who require NIV are managed on a respiratory ward (Tarka only as due to environmental challenges with Roborough is not a suitable space to support acute patients needing NIV/HFNO). In the event of ward closure options need to be discussed with the involvement of the infection control team
- Principles of safe staffing are embedded where supporting NIV patients at ward level, particularly in relation to the acute phase of care.

3. DEFINITIONS

AHRF - Acute Hypercapnic Respiratory Failure

NIV – Non Invasive Ventilation

EPAP – Expiratory Positive Airway Pressure

IPAP – Inspiratory Positive Airway Pressure

TEP – Treatment Escalation Plan

COPD – Chronic Obstructive Pulmonary Disease

OHS – Obesity Hypoventilation Syndrome

KS – Kyphoscoliosis

NMD – Neuromuscular disease

ABG – Arterial Blood Gas

HFNO- High Flow Nasal Oxygen

4. DUTIES AND RESPONSIBILITIES OF STAFF

4.1. Medical staff

The Doctor or senior advanced practitioner commencing NIV is responsible for:

- Identifying a patient has AHRF
- Ensuring the indication for NIV is correct
- Ensuring there are no contraindications to NIV
- Contacting the respiratory Ward nursing staff to inform them of patient
- Giving nursing staff guidance on treatment targets
- Ensuring a treatment escalation plan is documented prior to commencing NIV
- Reviewing treatment efficacy and advising treatment changes
- Referring patient to respiratory team within 2 hours of admission or at start of next working day.
- Considerations of Treatment Escalation Plan (TEP) must be made on each patient based on an individual circumstance and prior to discharge to the ward. For some patients this may be a ceiling of treatment however, for some patients it may be appropriate for them to be for full escalation and moved to an acute respiratory ward for ongoing management.

4.2. ITU referral

Consideration is required for patients who may be appropriate for ITU review. Patients with pH < 7.25, and PaCO2 > 6.0, escalation of care should be discussed with critical care at the time of or soon after commencement of NIV (NIV should not be delayed).

Patients who meet the critical care criteria for admission should be referred to the intensivist team

NIV may be considered as a ceiling of care for these patients, advice can be sought from the Respiratory Physicians (working hours 9am – 5pm) or from ITU.

4.3. Nursing Staff

The clinical environment for NIV should facilitate enhanced nursing. This should be in line with staffing levels associated with level 2 care (1:2) and appropriate monitoring facilities

The Nurses managing the NIV are responsible for:

- Ensuring they are familiar with the guidelines for managing hypercapnic respiratory failure
- Setting up the equipment
- Monitoring the patient
- Ensuring the ward is adequately staffed

- Ensuring they have undertaken a trust competency to be able to manage the patients care
- Being aware of the patient's escalation plan and understand that some patients will be transferred to an acute respiratory ward who are deemed appropriate for full escalation to ITU areas.

The same consideration and staffing ratios must also be achieved for patients who requires high flow nasal oxygen (HFNO) in the ward environment. Please see HFNO policy.

5. Documentation

5.1. NIV documentation is to be completed on EPIC under flow sheets and complex vitals and complete the ventilator settings. It is the responsibility of the nurse to complete this section to enable accurate record of NIV usage.

6. Indications for ward based NIV

6.1. The criteria for ward based NIV is shown in appendix 1

6.2. COPD

pH <7.35 **and**

pCO2 >6.5

After initial 1 hour period of optimal medical management including controlled oxygen therapy **unless** initial pH <7.25 in which case NIV should be started immediately

6.3. Neuromuscular Disease

Respiratory illness **+/-** RR >20 with known/suspected respiratory muscle weakness **or**

pH <7.35 and pCO2 >6.5

6.4. Obesity

pH <7.35, pCO2 >6.5 and RR >23 **or**

pCO2 >6.0 and daytime somnolence

7. Ward NIV not indicated – consider ITU referral

7.1. Asthma

7.2. Pneumonia- Consider high flow nasal oxygen

7.3. Other causes of AHRF eg drug overdose, imminent death

Where NIV has been commenced in the Emergency Department and diagnosis is not COPD, MND or OHS the patient must be referred to HDU by the emergency department for ongoing care

See flow diagram in appendix 2 for NIV indication

8. Contraindications to NIV

8.1. Absolute

- Respiratory arrest/peri-arrest
- Severe facial deformity
- Facial burns
- Fixed upper airways obstruction

8.2. Relative – will need enhanced observation, consider HDU

- Undrained pneumothorax
- GCS <8 where airway is compromised
- pH <7.15 (or <7.25 with other adverse features)
- Confusion/agitation (that would affect ability to co-operate with treatment)
- Cognitive impairment
- Uncontrollable vomiting
- If NIV is the ceiling of care it may be used where adverse factors are present. The patient will need close observation for complications of treatment.

9. Setting up NIV

NIV should only be started in ED, HDU or on an appropriate respiratory unit Following an ABG. NIV should not be commenced from a venous blood gas

The respiratory ward must be fully staffed to accept a patient for NIV(see appendix 2 for safe staffing requirments, escalations and actions). This enables the close monitoring of patients starting NIV in line with national guidelines.

If the respiratory Ward is understaffed patients requiring NIV may need to be admitted to HDU even if their ceiling of treatment is for ward based care. This is to enable safe nursing. A risk assessment should be undertaken by the medical team in consultation with ITU if this situation occurs.

When the respiratory Ward is closed patients on NIV may be managed on HDU or a secondary ward who can be supported by the respiratory ward staff. Duty matrons will need to consider skill mix and staff deployment in these situations.

A treatment escalation plan should be documented **prior** to commencing NIV

Ventilator setting are to be completed on EPIC complex vitals section

NIV should be started within 1 hour of the ABG confirming it is required

9.1. Mask

A full-face mask should be used as the first interface, sized appropriately to the patient

Masks for home NIV machines are not compatible with the in-patient machines and **must not** be used

Avoid over tightening of the mask. This can cause nasal ulceration which can lead to NIV failure

In relation to COVID 19 the machine must be turned off before the removal of any mask and the mask reapplied prior to switching the machine on

9.2. Other care

For COVID 19 the bacterial filter needs to be placed between the mask and the circuit (which is above the exhalation port)

Provide care to eyes and mouth.

Breaks from NIV should be made for drinks, drugs, physiotherapy, meals etc. Please refer to point 8.11 when mask is removed

Assess and monitor oral and dietary intake

All patients receiving NIV therapy should have a fluid balance chart to assist hydration assessment

NIV should be interrupted for the administration of nebulised bronchodilators

Administration of in line nebulisers via a standard nebuliser device is ineffective as the patient will only receive a small dose of the intended drug therapy

If the patient is unable to tolerate a break in NIV for nebuliser therapy continue NIV and seek senior advice.

Where an aerosol device is available (e.g. aeroneb in critical care) this may be used within the circuit as per manufacture guidelines.

9.3. Supplementary Oxygen

Supplemental oxygen can be added via port on the mask or when used with the trilogy oxygen is delivered through the machine and does not require additional oxygen tubing

Oxygen to be titrated to prescribed oxygen saturation of 88-92%

A humidified NIV system is not routinely used as this will affect the integrity of the bacteria filter when used with a dry circuit

9.4. Pressure settings

Initial EPAP: 3cmH2O (unless OSA when may be higher)

Initial IPAP: 15 (20 if pH <7.25)

Up-titrate IPAP over 30 min to target IPAP 20-30 cmH2O

Aim for pressure sufficient to augment chest movement and slow respiratory rate

Higher EPAP may be needed in severe OHS and chest wall deformity

9.5. Back up rate

A back up rate of 8-12 should be used

10. Monitoring patients on NIV

10.1. Patients should be on continual saturation monitoring whilst on NIV

10.2. Target saturations should be prescribed as 88-92%

10.3. Continuous ECG monitoring must be used for the first 12 hours of NIV treatment and continued if there is ongoing tachycardia (pulse >120) or hypotension (systolic BP <100)

10.4. NIV settings and use must be documented on epic under complex vitals and complete the ventilator section. This will support the clinician during the weaning process

10.5. ABG should be repeated to monitor pH and pCO2 within 1 hour of achieving target settings then again at 4 hours and 12 hours

10.6. Patient should use NIV as much as tolerated in the first 24 hours

10.7. If a patient is not responding to NIV consider

10.7.1. Synchronisation
10.7.2. Mask fit/leak
10.7.3. Exhalation port patency
10.7.4. Adequate treatment of underlying condition (physiotherapy/bronchodilators)
10.7.5. HDU/ITU escalation

11. Weaning from NIV

11.1. Weaning from NIV can commence once arterial blood gases return to acceptable range set by medical team

11.2. Assess readiness for weaning on a daily basis

11.3. Aim for normalisation of pH and pCO2

11.4. Patients with chronic hypercapnia may not achieve a normal pCO2

11.5. Taper use of daytime NIV with breaks of 2-3 hours, gradually increasing duration of breaks

11.6. Discontinue nocturnal NIV when self-ventilating pCO2 is within agreed target range

12. Decontamination and cleaning

The machine should be wiped over/ cleaned daily in accordance with Trust cleaning standards. Following use the machine should be cleaned and decontaminated following infection control and manufacturer's instructions.

There is a reusable foam inlet filter, required to protect the ventilator from dirt and dust. This requires cleaning every week and replacing periodically (recommended every 6 months) and/or if damaged.
In relation to COVID 19 no additional cleaning and filter changes needs to be made from the normal process

13. Monitoring Compliance with and the Effectiveness of the Guideline

13.1. Education

- All Staff in ED and the respiratory ward who have contact with NIV patients must receive NIV training. From the Advanced Clinical Practitioner NIV trust Lead All nursing staff to undertake a competency assessment

13.2. Process for Implementation and Monitoring Compliance and Effectiveness

This policy will be monitored and updated to ensure that best practice and new evidence will be considered

Details of monitoring

- Audit of compliance of NIV proforma completion
- Annual audit undertaken on the patients who require NIV with a patient register being completed
- Audit on the completion of the daily NIV observation chart

14. Patients on long term NIV

14.1 For patient who are on long term NIV at home and admitted to hospital due to another clinical reason, medical, surgical or orthopaedic. These patients DO NOT need to be managed on an acute respiratory ward and can be placed on other wards throughout the hospital as required.

15. ARCHIVING ARRANGEMENTS

The original of this SOP will remain with the author ACP Respiratory medicine. An electronic copy will be maintained on the Trust intranet. Archived electronic copies

will be stored on the Trust's "archived policies" shared drive, and will be held indefinitely. A paper copy (where one exists) will be retained for 10 years.

16. PROCESS FOR MONITORING COMPLIANCE WITH AND EFFECTIVENESS OF THE STANDARD OPERATING PROCEDURE/ GUIDELINE

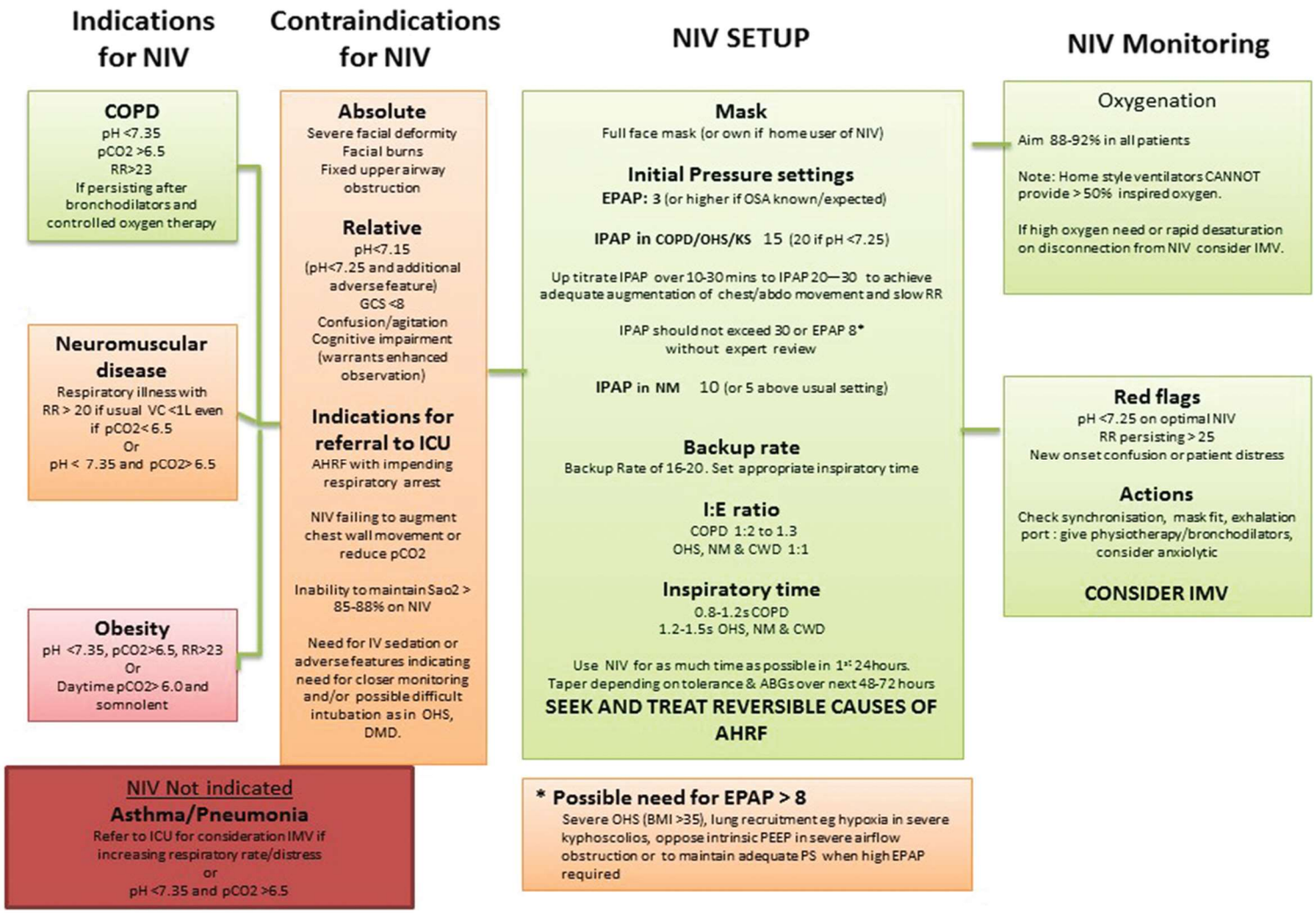
16.1. To evidence compliance with this policy, the following elements will be monitored:

What areas need to be monitored?	How will this be evidenced?	Where will this be reported and by whom?
75% of ward staff competent in NIV management	Matrons dashboard	Matrons governance meeting NDDH
Safe staffing (reportable red flags and <75% nursing hours)	Divisional governance and PAFs	Nursing Assurance and productivity group

17. REFERENCES

- BTS/ICS Guidelines for the Ventilatory Management of Acute Hypercapnic Respiratory Failure in Adults
- Inspiring Change – NCEPOD report into use of acute non-invasive ventilation

Appendix 1 NIV flow chart



Appendix 2 NIV competency assessment

	Nursing competencies	Assessment criteria	Nurses signature	Assessors signature	Date achieved
1	Demonstrate an understanding of the anatomy and physiology of the respiratory system	Identify the main structures of the respiratory system Discuss the position of these structures in relation to their function Discuss the mechanism of normal breathing by negative pressure Identify the difference between type 1 and type 2 respiratory failure and discuss why some patients may be oxygen sensitive and target saturations may need to be altered			
2	Identify the contraindications for NIV (BIPAP)	Be able to identify the contraindications for commencing NIV			
3	Correctly assemble and prepare the equipment	Ensure bipap machine is correctly plugged in and switched on Ensure filters are changed as required Identify and measure the patient for the correct mask Identify exhalation port on hose Check mask for leaks Check patients breathing with the machine			
4	Explain what is needed prior to starting NIV therapy, check capacity, consent and check for medical escalation plan	Check NIV prescription and proforma has been completed Any issues with consent/capacity? Ensure chest xray has been requested and completed to rule out pneumothorax (DO NOT DELAY TREATMENT WAITING FOR CXR) Ensure BIPAP is being used as per local policy			
5	Provide an explanation to the patient and to the next of kin	Provide advice to the patient on what will happen, the type of mask to be used and the effects of treatment			
6	Explain and implement the safety measure required for BiPAP	Monitoring of patient with cardiac monitor with continual saturation monitoring Oxygen therapy prescribed			

		Understand the alarms on the machine			
7	Demonstrate the ability to commence the treatment (on a patient) and plan within the first hour, with underpinning rationale for actions	Understanding of acute hypercapnia and type 2 respiratory failure Understanding of IPAP setting Understanding of EPAP settings Understand how to increase the setting for IPAP and EPAP parameters Demonstrate how to monitor a patients respiratory rate and o2 saturations Record accurately how to record an EWS			
8	Accurately record the treatment of the patient, changes in the patient's condition and maintains appropriate documentation throughout the patient' treatment	Able to demonstrate how to complete the EWS and when to escalate appropriately Demonstrate accurate documentation on the NIV daily chart Demonstrate clear documentation as per NMC guidelines. Demonstrate the ability to be able to clearly and accurately document an ABG			
9	Ensure medication is administered to the patient receiving BiPAP therapy and discuss actions to take if the patient desaturates on mask removal.	Demonstrate the understanding of using other therapies eg nebuliser (In line and air driven), steroids, use of antibiotics,			
10	Hydration & fluid balance	Show an understanding of how to ensure patients receive adequate hydration and nutrition while receiving bipap Explain how to facilitate breaks for fluid, nutrition and medication Identify how to record nutritional and fluid balance on these patients			
11	Complications with NIV	Discuss how you would manage non-compliance with treatment How would you manage patients coming to the end of life			
12	Maintaining of machines	What filters do you need to change? How often do they require changing Cleaning the machine			

Appendix 3 Safe staffing – escalation and actions

Nursing RAG ratings

Tarka/Roborough (28 beds)					
	Day		Night		
Shift	Green	RN: Pt	Night	RN: Pt	Comments
Green	6 RNs 5 HCA's		5 RNs 4 HCAs		Ward split over 2 separate footprints Tarka – 20 beds Roborough – 8 beds
Amber	5 RNs 4 HCAs		4 RNs 3 HCA's		Consideration to be given with acuity of patients as may change RAG rating if have high numbers of NIV
Red	4 RNs 2 HCA's		3 RNs 2 HCAs		

STAFFING GREEN

Capacity

4 NIV Patients

Nursing Split Tarka only

- 4 registrants on ward
- 2 registrants allocated to NIV (1:2)
- 2 registrants split between remaining 16 patients (1:8)

**additional Registrant to be consider at night to bring up to day time staffing levels*

STAFFING AMBER

Capacity

2 NIV patients

Nursing split Tarka only

- 3 registrants on ward
- 1 registrant allocated to NIV (1:2)
- 2 registrants split between remaining 18 patients (1:9)

**additional Registrant to be consider at night to bring up to day time staffing levels*

STAFFING RED

Capacity

Would not be able to achieve safe staffing for NIV

- Follow staffing escalation/mitigation plans for RED ward
- Escalate to duty matron/senior nurse
- Consider support from ICU

Additional nursing needed by escalating NIV patients (based on green day staffing levels)			
Number of NIV patients	**RAG**	**Additional staffing**	**Actions**
4 NIV patients		1 registrant night	• update safe care to reflect level 2 care needs • Flag NIV patients to divisional/duty matron • Review additional registrant requirements overnight - to be escalated through divisional matron/duty matron (*consider acuity and dependency of other inpatients*) • Review staffing for proceeding shift to ensure safe staffing **to meet demands**

5-6 NIV Patients		1 registrant day 2 registrants' night	• Update safe care to reflect level 2 care needs
			• Add red flag to safe care to indicate shortfall in nursing hours
			• Escalate shortfall to divisional matron/duty matron (inaction escalation plan/mitigating actions associated with RAG scoring)
			• Review additional registrant requirements overnight - to be escalated through divisional matron/duty matron (*consider acuity and dependency of other inpatients*)
			• Review staffing for proceeding shift to ensure safe staffing **to meet demands**
			• At times of RED consider the following
			  ○ Critical care support
			  ○ Utilisation of other available resources (AHP, CNS, ACP)

>7		2 registrants' day and 3 registrants' night *Further escalation would be required with >10 patients*	

APPENDIX 5: EQUALITY IMPACT ASSESSMENT TOOL

Name of document	Standard Operating Procedure for managing hypercapnic respiratory failure with Non-Invasive Ventilation (Northern Services)
Division/Directorate and service area	Medical Division
Name, job title and contact details of person completing the assessment	Lucy Bates, Associate Director Nursing
Date completed:	18/5/23

The purpose of this tool is to:

- **identify** the equality issues related to a policy, procedure or strategy
- **summarise the work done** during the development of the document to reduce negative impacts or to maximise benefit
- **highlight unresolved issues** with the policy/procedure/strategy which cannot be removed but which will be monitored, and set out how this will be done.

1. **What is the main purpose of this document?**
   To detail processes and support required when patients are requiring NIV

2. **Who does it mainly affect?** *(Please insert an "x" as appropriate:)*

   Carers ☐ Staff ☒ Patients ☐ Other (please specify)

3. **Who might the policy have a 'differential' effect on, considering the "protected characteristics" below?** (By *differential* we mean, for example that a policy may have a noticeably more positive or negative impact on a particular group e.g. it may be more beneficial for women than for men)
   Please insert an "x" in the appropriate box *(x)*

Protected characteristic	Relevant	Not relevant
Age	X	
Disability	☐	X
Sex - *including: Transgender, and Pregnancy / Maternity*	☐	X
Race	☐	X

Religion / belief	☐	X
Sexual orientation – *including: Marriage / Civil Partnership*	☐	X

4. Apart from those with protected characteristics, which other groups in society might this document be particularly relevant to… (e.g. those affected by homelessness, bariatric patients, end of life patients, those with carers etc.)?

Any person who requires NIV (excluding children)

5. Do you think the document meets our human rights obligations? ☒

Feel free to expand on any human rights considerations in question 6 below.

A quick guide to human rights:
• ***Fairness –*** *how have you made sure it treats everyone justly?* • ***Respect –*** *how have you made sure it respects everyone as a person?* • ***Equality –*** *how does it give everyone an equal chance to get whatever it is offering?* • ***Dignity –*** *have you made sure it treats everyone with dignity?* • ***Autonomy –*** *Does it enable people to make decisions for themselves?*

6. Looking back at questions 3, 4 and 5, can you summarise what has been done during the production of this document and your consultation process to support our equality / human rights / inclusion commitments?

The document related to adults and excluded children. This is associated with differing environments and setting required for the more specialist paediatric patients.

7. If you have noted any 'missed opportunities', or perhaps noted that there remains some concern about a potentially negative impact please note this below and how this will be monitored/addressed.

"Protected characteristic":	
Issue:	
How is this going to be monitored/ addressed in the future:	
Group that will be responsible for ensuring this carried out:	